

SMARTBRIDGE

Let's Bridge the Gap | SKILL WALLET - A SmartBridge Product

Date	29-09-2026
Team ID	SWTID-2026-7949
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value

Scalability & Future Plan

Scalability & Future Plan:
This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High/Medium/Low)
1	Static Branch Evaluation in ACL Script: The current ACL condition specifically checks 'Branch is EEE' using static string comparison.	Adding new departments (e.g., Mechanical, Civil) requires configuring new discrete ACL records manually.	High
2	Direct User-to-Role Assignment: Roles bb1 to bb4 are mapped directly to individual user accounts rather than via ServiceNow User Groups.	Administrative overhead increases significantly when provisioning hundreds of departmental faculty members.	Medium
3	Standard List View Count Message: The list header indicates '6 records matching' while only displaying the 2 authorized EEE rows to the end user.	May cause minor visual confusion for non-technical academic staff, though security isolation remains 100% airtight.	Low

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Individual test user accounts with direct role assignments (bb1 to bb4).	Migrate to ServiceNow Group-Based Role Assignment (sys_user_grmember), syncing automatically with university SSO/LDAP.
2	Data Storage	Custom table 'u_institution_details' storing sample records in a single database partition.	Implement Table Partitioning, Table Archive Rules, and composite indexing on (Branch,

			sys_id) for 100k+ records.
3	Performance	Direct server-side JavaScript evaluation during each table query.	Deploy Script Includes with cached role lookups, reducing server query overhead from <80ms to <20ms under heavy concurrency.
4	Security	Static ACL rule matching 'Branch is EEE'.	Implement Dynamic Script-Controlled ACLs matching user's department profile dynamically (current.branch == gs.getUser().getDepartment()).

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Multi-Department Dynamic Segregation: Replace static branch matching with dynamic user-department comparison across all university departments (ECE, CSE, MECH, CIVIL).	Q4 2026	Eliminates redundant ACL rules; zero maintenance when onboarding new academic branches.
Phase 3	Service Portal Integration & Custom Widget Security: Implement secured Service Portal widgets using GlideRecordSecure and REST Table API authentication.	Q1 2027	Enables self-service student/faculty portal with guaranteed zero cross-branch data leaks.
Phase 4	Automated Security Audit & Compliance Analytics: Implement automated scheduled jobs and Performance Analytics dashboards tracking ACL evaluation traces.	Q2 2027	Provides real-time compliance reporting for institutional accreditation and FERPA audits.